Iron-Catalyzed Exhaustive Reduction of Esters under Low H_2 Pressures Enabled by Magnetocatalysis

Sihana Ahmedi,[a,b] *Walter Leitner,*[a,b] *Alexis Bordet*[a*]

[a]*Max Planck Institute for Chemical Energy Conversion, Mülheim an der Ruhr, Germany*

[b]*Institute of Technical and Macromolecular Chemistry RWTH Aachen University, Aachen, Germany*

The exhaustive reduction of esters all the way to methyl-containing compounds is challenging but of fundamental importance to enable degradation and recycling of polyesters, as well as to synthesize methyl and tolyl derivatives in synthetic chemistry.[1-2] However, this transformation is typically carried out following non-sustainable pathways employing excess of strong and often hazardous reducing agents (e.g. metal hydrides, hydrosilanes, ammonia borane) under harsh conditions.[2-3] As a result, the development of catalytic and atom-economical ester reduction pathways using H_2 as a reducing agent under mild conditions is highly desired, but remains so far a challenge.

In this context, we explore in the present study the possibility to achieve selective ester reduction with H_2 using iron-based nanoparticles (NPs) with magnetic heating capabilities.[4] Indeed, magnetic induction heating has been demonstrated by our group and others to heat suitable catalyst materials in a very rapid, localized, and energy efficient manner.[5-6] In particular, this technology obviates the necessity of heating the entire reactor environment, potentially leading to lower overall process temperatures and reduced energy consumption. In the present work, we prepare and characterize iron carbide nanoparticles (ICNPs) with exceptional induction heating capabilities, and apply them to the magnetocatalytic reduction of esters using H_2.

ICNPs were prepared following an organometallic approach,[7] and characterized by various techniques including electron microscopy, XRD, Mössbauer spectroscopy, vibrating sample magnetometry. Activated by magnetic induction heating, ICNPs exhibited high activity, selectivity, and stability for the reduction of a large scope of esters (including PET models) under mild conditions (3 bar H_2, 200 °C).[4] Aromaticity was maintained in case of benzylic ester substrates. Strikingly, ICNPs heated conventionally at 200 °C showed no activity. The catalyst's adaptability to intermittent power supply was demonstrated by regularly stopping and re-starting the AC magnetic field. This work may pave the way toward a greatly enhanced practical implementation

of ester reduction in research and industry, and demonstrates more generally the broad potential of the emerging field of magnetocatalysis for synthetic transformations typically requiring demanding operating conditions.

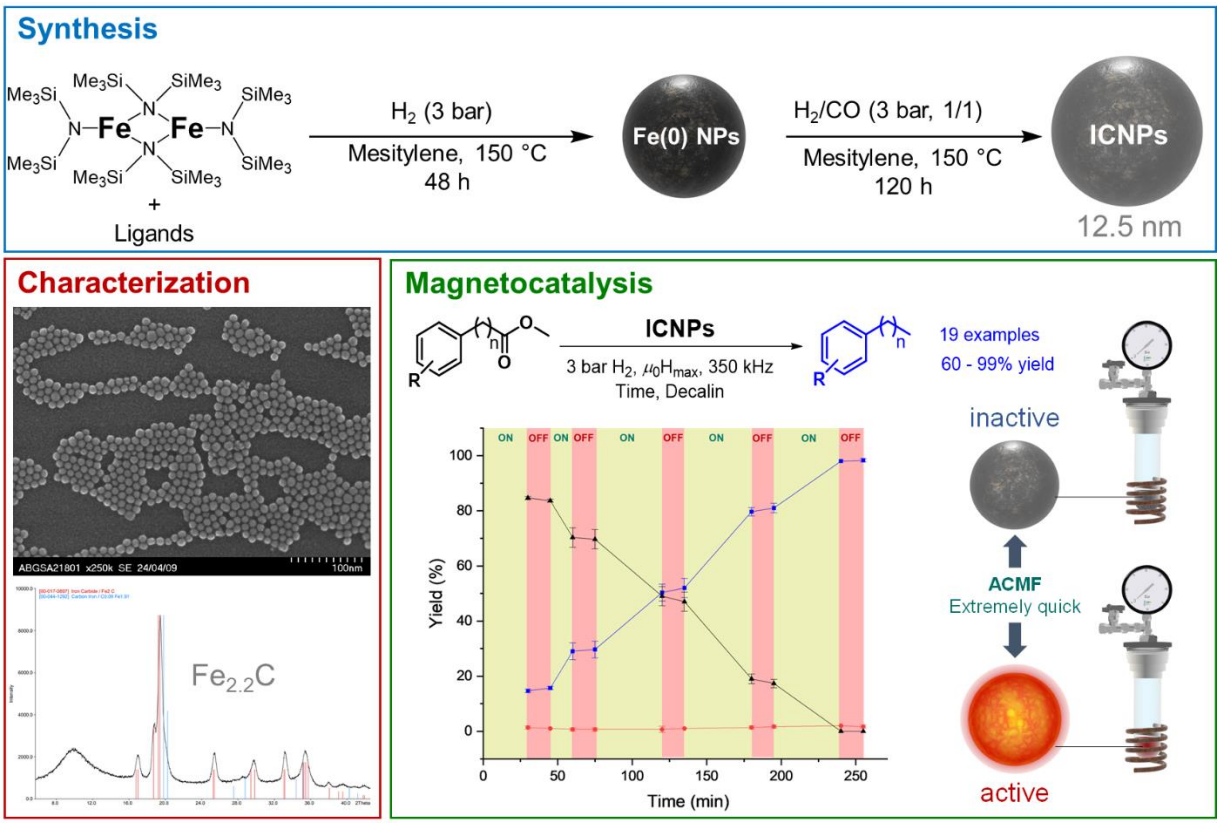

Figure 1. Illustration of the approach, with the synthesis of ICNPs, their characterization, and their application to the magnetocatalytic reduction of esters.

References

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